Mid Term: Characterization of Materials (MT205(TO))

भारतीय प्रौद्योगिकी संस्थान जम्मू
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Mid-Term
COURSE: Characterization of Materials

Total time: 60 mins

Answer All the Questions (Total = 30)

1. Observe the TEM diffraction pattern carefully. The mixed indices of the planes (one of them marked with an arrow) with lower intensity are halfway lighted up in between the fundamental planes with even indices (one of them shown as $02\bar{2}$) in the reciprocal space. (a) Complete the indexing with the provided information. (b) Find out the zone axis of the diffraction pattern. (c) Can you suggest the crystal structure of this binary alloy system? If not then devise a recipe for the proper identification of the crystal structure with required schematic. (d) Now, from your understanding, suggest some planes which could be visible at [112] zone axis. (e) Construct the **Unit Cell** for the binary alloy system taking **both even and mixed indices planes in a same coordinate system.** (f) Devise a probabilistic **Structure Factor Scheme** of the binary alloy system.

[2 2̄ 2̄] → SC

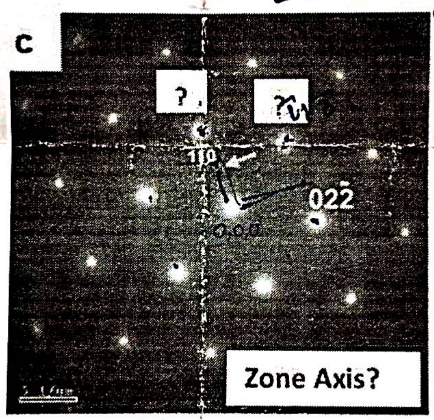
2̄ 2̄ 0   1̄ 1̄ 0   0 2 2̄

(2+2+3+2+5+6 = 20)

1̄ 1 2̄

2̄ 2 0

2̄ 2 2̄



y, x

2. The translational vector of an FCC lattice can be written in the Equation Set 1: (Where "a" is the edge length). Now if the corresponding reciprocal translation vector can be expressed as Equation Set 2. **Then prove that a real FCC lattice becomes BCC lattice in the reciprocal space.** Find out the edge length of the formed BCC lattice in the reciprocal space. **(10)**

$$\boxed{\begin{aligned}\vec{a}_1 &= \frac{a}{2}\cdot(\hat{y}+\hat{z})\\ \vec{a}_2 &= \frac{a}{2}\cdot(\hat{x}+\hat{z})\\ \vec{a}_3 &= \frac{a}{2}\cdot(\hat{x}+\hat{y})\end{aligned}}$$

.... **Equation Set-1**

Here $\hat{x}$, $\hat{y}$ and $\hat{z}$ denote the unit vectors in x-, y-, and z direction.

with $V = \vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)$

$$\boxed{\begin{aligned}\vec{b}_1 &= 2\pi\cdot\frac{\vec{a}_2\times\vec{a}_3}{V}\\ \vec{b}_2 &= 2\pi\cdot\frac{\vec{a}_3\times\vec{a}_1}{V}\\ \vec{b}_3 &= 2\pi\cdot\frac{\vec{a}_1\times\vec{a}_2}{V}\end{aligned}}$$

.... **Equation Set-2**